



NATIONAL SCHOOL OF BUSINESS MANAGEMENT

Mid Term Exam

CS402.3 – Computer Graphics and Visualization

Instructions to Candidates

- 1) Answer all questions.**
- 2) Time allocated for the examination is three (03) hours
- 3) Total number of pages - Four (04)
- 4) Must use the answer script as the front page
- 5) No hand-drawn images should be used **except for question 04**
- 6) Submissions are accepted through NLEARN only
- 7) Late submissions are subjected to mark capping

Question 1 (Image Processing Fundamentals)**[25 marks]**

- 1) Take a look at the following blob tracking system. One can convert the image taken from a CCTV camera to identification of connected blobs as shown in the image on the right.



- a. Explain how you would convert colour image to binary image. You need to provide step by step process while explaining the steps. (15 marks)
- b. Explain two types of thresholding techniques you can use related to the original image. (10 marks)

0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0	0
0	0	0	1	1	1	0	0	0	0
0	0	0	1	0	1	0	0	0	0
0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0	0	0

- c. Take a look at the extracted blob. Show how you can confirm all of them belong to a single blob. (5 marks)

Question 2 (Histograms)**[25 marks]**

- 1) Given an image (left 5x5 sub image) as below, derive the intensity mapping that will as best as possible equalize the image histogram. The image histogram is as shown on the right. There are 8 possible grey scale levels from 0 to 7.

3	3	3	4	4
5	4	5	4	4
4	3	3	4	3
4	5	3	3	4
4	4	3	5	3

I	0	1	2	3	4	5	6	7
f(I)								
Cuf								
Feq								
CuFeq								

- Complete the above table by computing the intensities (I), intensity frequencies [$f(I)$], cumulative frequencies (Cuf), etc. (15 marks)
- Compute mappings; For each intensity in the original image, find an intensity in the transformed image that has as close as possible, the same amount of cumulative frequency. Describe how would you map the intensities for each and every input I and output O . (10 marks)
- Draw the equalized resulting 5x5 image. (5 marks)

Question 3 (Morphology and Filters)

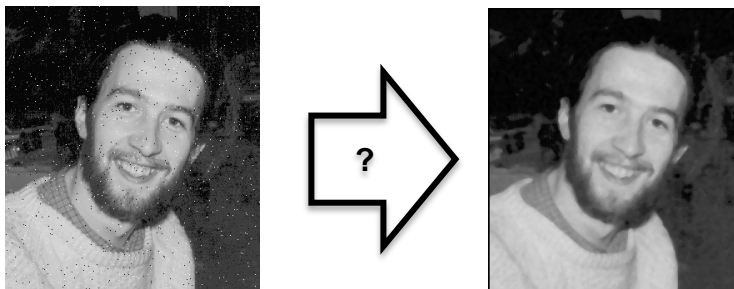
[25 marks]

- What is *noise* in signal processing and briefly explain how it is added to an image? (8 marks)
- Identify the following filter (H) and explain how you would get the value in the red box in the image G . (9 marks)

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} * F = G$$

H F G

- Suppose you are given the following image with *salt-and-pepper* noise. Explain how you would recover/enhance the image. (8 marks)



Question 4 (Hough Transformation)

[25 marks]

- Suppose image has 5 points: (1,0), (1,1), (1,2), (2,2) and (3,3). Each of these points corresponds to the following lines that can be plotted. Using the Hough transformation show that those points are collinear and find the line equation(s). [You are free to choose between parameter space or HNF] (25 marks)

